Dashboard / Meine Kurse / 2023S365249 / Final Exam 29th June -- 17:15 -- 21:00 / [Exam] PG 2

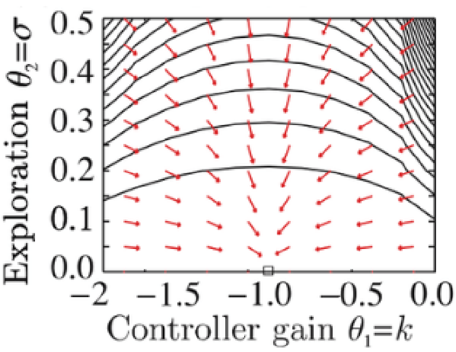
Begonnen am	Thursday, 20. April 2023, 17:05
Status	Beendet
Beendet am	Thursday, 20. April 2023, 17:08
Verbrauchte Zeit	2 Minuten 45 Sekunden
Bewertung	**3,50** von 10,00 (**35**%)

Frage **1**

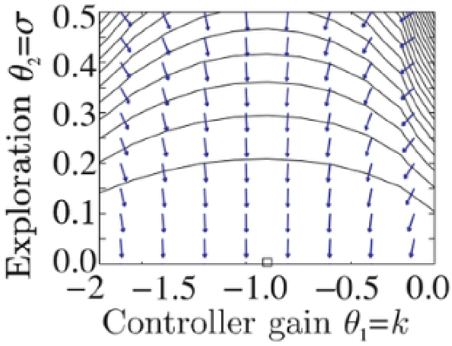
Vollständig

Erreichte Punkte 0,00 von 1,00

Match the plots with their respective method



Policy Gradient with euclidean distance



Natural Policy Gradient

Frage **2**

Vollständig

Erreichte Punkte 0,50 von 1,00

Explain the main differences between Proximal Policy Optimization (PPO) and Trust Region Policy Optimization (TRPO). How does PPO address some of the limitations of TRPO and what are the benefits of using PPO?

- [x] a. PPO and TRPO both use a trust region constraint to ensure stability, but PPO uses a clipped objective function instead of a KL divergence constraint.
- [ ] b. PPO is a more scalable and practical alternative to TRPO, as it uses a simpler and more computationally efficient objective function.
- [ ] c. PPO and TRPO have similar computational complexity, but PPO achieves better sample efficiency and convergence properties.
- [ ] d. PPO and TRPO are equivalent methods that differ only in their implementation details.

Frage **3**

Vollständig

Erreichte Punkte 0,00 von 1,00

Suppose you are implementing TRPO with a neural network policy that outputs a Gaussian distribution over actions. Which of the following statements is true regarding the KL-divergence constraint in TRPO?

- [ ] a. The KL-divergence between the old and new policy distributions should be kept below a certain threshold.
- [x] b. The KL-divergence constraint is not used in TRPO when the policy outputs a Gaussian distribution.
- [ ] c. The KL-divergence between the old and new policy distributions should be minimized as much as possible.
- [ ] d. The KL-divergence between the old and new policy distributions should be exactly equal to a certain threshold.

Frage **4**

Vollständig

Erreichte Punkte 0,00 von 1,00

Which of the following statements is true regarding Trust Region Policy Optimization (TRPO)?

- [ ] a. TRPO uses a constraint on the maximum change in policy parameters to ensure stability and monotonic improvement.
- [ ] b. TRPO is only applicable to continuous action spaces.
- [x] c. TRPO is a type of off-policy algorithm that updates the policy directly based on the gradient of the expected reward with respect to the policy parameters.
- [ ] d. TRPO is a type of off-policy algorithm that learns the value function of a different policy.

Frage **5**

Vollständig

Erreichte Punkte 1,00 von 1,00

Which of the following statements is true regarding Proximal Policy Optimization (PPO)?

- [ ] a. PPO is an off-policy algorithm that learns the value function of a different policy.
- [ ] b. PPO updates the policy parameters using the natural gradient of the expected reward with respect to the policy parameters.
- [ ] c. PPO is only applicable to continuous action spaces.
- [x] d. PPO uses a clipping objective function that constrains the change in the policy probabilities between updates to ensure stability and monotonic improvement.

Frage **6**

Vollständig

Erreichte Punkte 0,00 von 1,00

What is the policy improvement in policy gradient, as seen in class?

Wählen Sie eine oder mehrere Antworten:

- [x] a. $J(\theta') - J(\theta) = E_{\tau \sim p_\theta(\tau)} \left[\sum_t \gamma^t Q^{\pi_\theta}(s_t, a_t) \right]$
- [ ] b. $J(\theta') - J(\theta) = E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_t \gamma^t A^{\pi_\theta}(s_t, a_t) \right]$
- [ ] c. $J(\theta') - J(\theta) = E_{\tau \sim p_{\theta'}(\tau)} \left[\sum_t \gamma^t Q^{\pi_\theta}(s_t, a_t) \right]$
- [ ] d. $J(\theta') - J(\theta) = E_{\tau \sim p_\theta(\tau)} \left[\sum_t \gamma^t A^{\pi_\theta}(s_t, a_t) \right]$

Frage **7**

Vollständig

Erreichte Punkte 0,50 von 1,00

Can we approximate this expression

$\sum_t E_{s_t \sim p_{\theta'}(s_t)} \left[E_{a_t \sim \pi_\theta}(a_t \mid s_t) \left[\gamma^t \frac{\pi_{\theta'}(a_t|s_t)}{\pi_\theta(a_t|s_t)} A^{\pi_\theta}(s_t, a_t) \right] \right]$

with this one:

$\sum_t E_{s_t \sim p_\theta(s_t)} \left[E_{a_t \sim \pi_\theta}(a_t \mid s_t) \left[\gamma^t \frac{\pi_{\theta'}(a_t|s_t)}{\pi_\theta(a_t|s_t)} A^{\pi_\theta}(s_t, a_t) \right] \right]$

And therefore, ignore the distribution mismatch?

(the only difference between these two expressions is just the outer expectation. $E_{s_t \sim p_{\theta'}(s_t)}$ and $E_{s_t \sim p_\theta(s_t)}$)

Wählen Sie eine oder mehrere Antworten:

- [ ] a. No. This approximation only makes sense in deterministics environments.
- [ ] b. No. This approximation is not possible.
- [x] c. Yes, and the error in the approximation will depend on the trajectory distribution mismatch.
- [ ] d. Yes, and the error in the approximation will depend on the policy.

Frage **8**

Vollständig

Erreichte Punkte 0,00 von 1,00

Assume that π_θ, and π'_θ are both an arbitrary distribution and that $\pi_{\theta'}$ is close to π_θ if $|\pi_{\theta'}(a_t \mid s_t) - \pi_\theta(a_t \mid s_t)| \leq \epsilon, \forall s_t \, a_t$.

What is the state **distribution change** bound?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. $|p_{\theta'}(s_t) - p_\theta(s_t)| \leq 2\epsilon t$
- [x] b. $E_{p_{\theta'}(s_t)}[f(s_t)] \leq E_{p_\theta(s_t)} f(s_t) - 2\epsilon t \max_{s_t} f(s_t)$
- [ ] c. $|p_{\theta'}(s_t) - p_\theta(s_t)| \geq 2\epsilon t$
- [ ] d. $E_{p_{\theta'}(s_t)}[f(s_t)] \geq E_{p_\theta(s_t)} f(s_t) - 2\epsilon t \max_{s_t} f(s_t)$

Frage **9**

Vollständig

Erreichte Punkte 1,00 von 1,00

Assume that π_θ, and π'_θ are both an arbitrary distribution and that $\pi_{\theta'}$ is close to π_θ if $|\pi_{\theta'}(a_t \mid s_t) - \pi_\theta(a_t \mid s_t)| \leq \epsilon, \forall s_t\ a_t$.

What is the **objective value** bound?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. $E_{p_{\theta'}(s_t)}[f(s_t)] \leq E_{p_\theta(s_t)} f(s_t) - 2\epsilon t \max_{s_t} f(s_t)$
- [ ] b. $|p_{\theta'}(s_t) - p_\theta(s_t)| \leq 2\epsilon t$
- [ ] c. $|p_{\theta'}(s_t) - p_\theta(s_t)| \geq 2\epsilon t$
- [x] d. $E_{p_{\theta'}(s_t)}[f(s_t)] \geq E_{p_\theta(s_t)} f(s_t) - 2\epsilon t \max_{s_t} f(s_t)$

Frage **10**

Vollständig

Erreichte Punkte 0,50 von 1,00

Which of the following implementations correspond to the family of Proximal Policy Optimization methods?

$\rho_t = \frac{\pi_{\theta'}(a_t|s_t)}{\pi_\theta(a_t,|s_t)}$

$\rho_{clip} = clip(\rho_t, 1-\epsilon, 1+\epsilon)$

Wählen Sie eine oder mehrere Antworten:

- [ ] a. $\theta' \leftarrow argmax_{\theta'}\ E_\theta \left[A^{\pi_\theta}(s,a) \right]$
- [ ] b. $\theta' \leftarrow argmax_{\theta'}\ E_\theta \left[\rho_t A^{\pi_\theta}(s,a) - \beta KL(\pi_\theta, \pi_{\theta'}) \right]$
- [x] c. $\theta' \leftarrow argmax_{\theta'}\ E_\theta \left[\min\{\rho_t A^{\pi_\theta}(s,a), \rho_{clip} A^{\pi_\theta}(s,a)\} \right]$
- [ ] d. $\theta' \leftarrow argmax_{\theta'}\ E_\theta \left[A^{\pi_\theta}(s,a) - \beta KL(\pi_\theta, \pi_{\theta'}) \right]$
- [ ] e. $\theta' \leftarrow argmax_{\theta'}\ E_\theta \left[\min\{\rho_t, \rho_{clip}\} \right]$

◄ [Exam] PG 1

Direkt zu:

[Exam] Value Based Methods ►